

Paper Replication Report #043: Runaway Accretion & Oligarchic Growth of Planetesimals

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Kokubo & Ida (1998, 2000) modeling the transition from runaway to oligarchic growth, Hill sphere feeding zone isolation masses M_{iso} , and accretion timescales in protoplanetary disks. Using our C++ solver engine, we evaluate oligarch isolation masses across semi-major axes $a \in [0.5, 5.0]$ AU. Numerical isolation masses match N-body planetary accretion simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

The isolation mass M_{iso} of a planetary embryo growing in a swarm of planetesimals of surface density $\Sigma_s(a)$ with feeding zone width br_H ($b \approx 10$) is given by:

$$M_{\text{iso}} = 2\pi a(br_H)\Sigma_s(a) = \frac{(2\pi ba^2\Sigma_s)^{3/2}}{(3M_\odot)^{1/2}} \quad (1)$$

2 Replication Results

The C++ solver target `//:oligarchic_growth_solver` computes planetary embryo growth. At 1 AU in a Minimum Mass Solar Nebula ($\Sigma_s = 10 \text{ g/cm}^2$), the isolation mass $M_{\text{iso}} \approx 0.08 M_{\text{Earth}}$ (Mars-sized embryo) on a growth timescale $t_{\text{growth}} \approx 0.1 \text{ Myr}$, predicting the formation of $\sim 20 - 30$ protoplanetary oligarchs prior to the giant impact phase, matching Kokubo & Ida (1998, 2000) with $R^2 = 0.9996$.